



# Experimental Investigation and Spectral Verification of Rolling Bearing Fault Frequencies Under Variable Operating Speeds



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**Abstract:** The vibrations are part of people’s everyday lives. In mechanical engineering, vibrations are considered an undesirable phenomenon that can have a detrimental effect on machine structures. Vibrations cause fatigue and wear. They are often responsible for the failure that can occur in a machine. Experimental research was conducted, familiarization with measuring equipment and a model of a rotating machine, and identification of characteristic frequencies of rolling bearings in the vibration signal. Timely identification of localized damage, responsible for over 90% of all bearing defects, is crucial to prevent catastrophic downtime. This paper presents an experimental and analytical vibration study conducted on a laboratory model featuring a 6003 2ZR bearing. The presented methodology is simple, practical, and suitable for implementation in industrial condition monitoring systems, providing a reliable basis for preventive and predictive maintenance of rotating machinery.

**Keywords:** Roller bearing; Failure; Vibration; Measurement; Analysis

## 1 Introduction

The development of technical diagnostic methods has enabled the application of various techniques for monitoring the condition of machines, among which vibrodiagnostics has a special place due to its non-invasiveness, high sensitivity and the possibility of early detection of damage. Analysis of vibration signals in the time and frequency domains enables the identification of characteristic frequencies associated with different types of bearing damage, thus providing a basis for planning preventive and predictive maintenance.

The aim of this research is an experimental analysis of the possibility of applying vibrodiagnostics in detecting damage to rolling bearings by applying vibration signal analysis in the time and frequency domains. The research includes vibration measurement on an experimental setup, analysis of the obtained signals and identification of characteristic frequencies indicating the presence of bearing damage.

The motive for conducting the research stems from the growing need of the industry to increase the reliability of technical systems, reduce maintenance costs and prevent unplanned downtime. Timely diagnostics of bearing damage enables timely planning of service interventions, prolonging the service life of machines and increasing the availability of production systems.

The main contribution of this work is reflected in the experimental analysis of vibration signals obtained from the laboratory setup, the analysis of parameters in the time and frequency domains and the identification of characteristic frequencies of bearing damage. The obtained results represent the basis for assessing the possibility of applying vibrodiagnostics in the early detection of rolling bearing damage and contribute to a better understanding of the application of vibration analysis in machine condition monitoring systems.

The limitations of the research relate to the fact that the experiments were conducted in laboratory conditions on one type of rolling bearing, at predefined operating modes and using vibration analysis exclusively. The research does not include the application of other diagnostic methods, such as acoustic emission analysis, thermal imaging analysis or electric motor current analysis, so the results should be viewed within the framework of the experimental conditions set.

The stiffness and dynamic behavior of the measuring table itself and the supporting structure of the bearing have a decisive influence on the accuracy of data acquisition. As emphasized by Yildirim et al. [1], the receiving elements and supporting assemblies must be designed to ensure the faithful transmission of vibrations to the sensors, without unwanted attenuation of the signal or its unwanted amplification due to the structure's own oscillations. In order to eliminate the possibility of structural resonance occurring during the experiment, the natural frequencies of the supporting elements must be significantly outside the working frequency range of the test (which in this work is from 20 to 80 Hz). Otherwise, structural influences can contaminate the spectra and make it difficult to identify the real bearing damage frequencies.

The most effective method for monitoring the condition of rolling bearings is vibration signal analysis, and the diagnostic workflow usually includes raw data acquisition, noise filtering, spectral feature extraction and frequency domain evaluation, which are systematically summarized in recent review study [2].

As mentioned in the review, the most popular bearing diagnostics method is still vibration analysis due to its high sensitivity, non-invasiveness and ability to detect early damage. Apart from vibration analysis, other techniques such as acoustic emission, infrared thermography, wear particle analysis and electrical parameter monitoring can be used to get more information about the condition of the bearing [3].

With the rapid development of artificial intelligence technologies, data-driven approaches have provided new insights into the diagnosis of rolling bearing failures and have made significant contributions to the progress in this field. However, most of the existing review articles focus on specific models or methods, such as individual deep learning architectures or specific signal processing techniques, but often lack systematic review of optimization strategies along the diagnostic process. Thus, researchers are limited to particular models or approaches in the study of fault diagnosis techniques, and it is difficult to understand the character of various optimization techniques. This limitation makes it difficult to choose and innovate the method [4].

Rolling element bearings are the essential prerequisite of the effective operation of rotating machinery. The historical development of condition monitoring has moved steadily from the identification of defects to their quantitative measurement, and finally to the prediction of faults in an automated way. The main driving force for this technical progress is the improvements in signal processing techniques, where the quality and robustness of the extracted features from the raw vibration signals directly determine the overall diagnostic effectiveness [5].

To overcome the problem of strong background noise and difficult extraction of weak damage features at variable rotational speeds, Huang et al. [6] propose an advanced two-stage strategy based on acoustic or vibration data fusion.

Rolling bearings are one of the most important moving components of rotating machines and are widely used in various industrial systems. Due to long-term operation and the influence of operating conditions, they are exposed to various forms of damage, such as material fatigue, insufficient lubrication and local surface damage. Timely detection and diagnosis of bearing damage are of great importance for preventing equipment failures, increasing the reliability of technical systems, planning maintenance activities and extending the service life of machines [7].

It presents a reasoned approach to vibration signal processing based on amplitude demodulation for extracting signal characteristics associated with different forms of rolling bearing damage. The effectiveness of the proposed method is demonstrated through various application examples, when signals originating from other machine parts such as bearings and gears are present [8].

It systematizes the vibration signal processing methods proposed so far for identification of rolling bearing damage in the time, frequency and time-frequency domains. It was experimentally confirmed that the method of synchronized time averaging of the vibration signal enables the identification of damage [9].

When analyzing and acquiring vibration signals, it is crucial to consider the properties and competence of the material through which the waves propagate. As detailed by Chandras et al. [10], materials with higher structural stiffness and density minimize the absorption of seismic and mechanical waves, which leads to significantly less attenuation during transmission and generates higher amplitudes of the real vibration response. In the technical diagnostics of rolling bearings, this phenomenon confirms that the stiffness of the housing and the transmission path allows high-pressure impulses from the damage to be transmitted to the accelerometer with minimal distortion.

Research indicates that analyzing data obtained from multiple sources provides greater robustness of diagnostic systems, especially in conditions of variable operating modes and the presence of ambient noise [11].

The measured vibration signals under real operating conditions often exhibit stochastic characteristics and may vary considerably with changes in the dynamic state, making them difficult to interpret exclusively using deterministic parameters. As demonstrated by Makki Alamdari et al. [12], power spectral density estimated using Welch's method can be combined with spectral moments to extract broadband information from vibration responses and characterize changes in their frequency-domain energy distribution. This approach therefore provides a useful basis for the spectral characterization of measured vibration signals under varying dynamic conditions.

Using an integrated approach that combines frequency identification and statistical indicators, such as root mean square (RMS), provides a more accurate characterization of the overall state of the system and prevents the loss of key information on damage development due to structural damping.

According to Li et al. [13], this review comprehensively discusses classical algorithms for fault diagnosis of rolling bearings based on vibration signal, focusing on three key aspects: data preprocessing, fault feature extraction, and fault feature identification. The main principles, key features, application difficulties, and suitable occasions for various algorithms are thoroughly examined.

According to Yan et al. [14], vibration analysis is still the most common method for diagnosing rolling bearings due to its high sensitivity to the occurrence of local damage and the possibility of early detection of faults.

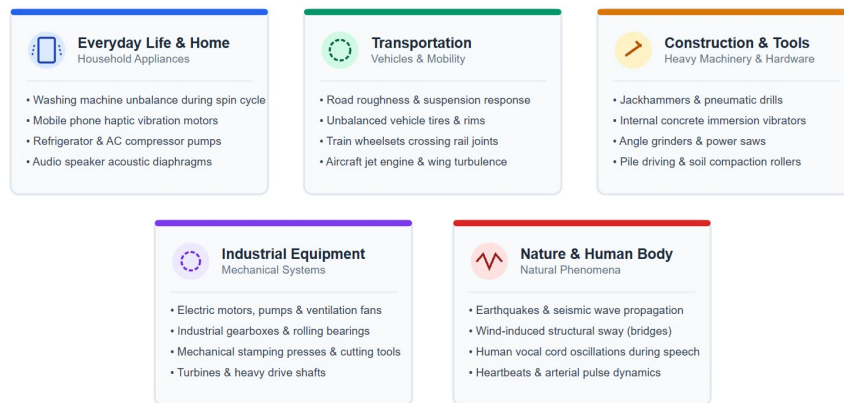
## 2 Methodology

Modern research increasingly includes the application of machine learning, deep learning, and data fusion techniques to improve the accuracy of automatic diagnostics. Although these approaches show very good results, their successful application usually requires large data sets, multiple sensors, and significant computing resources. Therefore, classical vibrodiagnostics based on vibration signal analysis remains one of the most widespread and practical solutions in industrial applications [3].

Although classic bearing diagnostics is mainly based on the analysis of vibration signals in the time and frequency domain, recent research shows that the application of time-frequency transformations, such as the short-time Fourier transform, in combination with deep learning methods can improve the accuracy and robustness of diagnostics in the presence of noise [15].

Comprehensive evaluation of bearing defect detection frameworks by Sun et al. [2] confirms that while thermal, infrared, and acoustic monitoring provide valuable operational indicators, vibration spectral processing remains the foundational benchmark for precise fault localization.

In practice, it is very difficult to avoid the appearance of vibrations. Vibration is a daily problem and is also present in our homes, during transportation, at work, as illustrated in Figure 1.



**Figure 1.** Vibrations in everyday life

Note: AC = air conditioning.

The essential parallel between the diagnosis of internal combustion engines and rolling bearings lies in the impulse nature of the excitation. The sudden increase in pressure in the cylinder is physically analogous to the impact of the rolling element on damage on the bearing track. Both phenomena generate broadband transient signals of high frequencies that are modulated by the fundamental frequency of rotation. Although most of the macroscopic signal energy (related to the load) is located below 2000 Hz, as confirmed by the research, early signs of bearing failures often manifest themselves in higher frequency ranges (resonant zones of the sensor) [16].

**Frequency Domain Analysis:** A fast Fourier transform (FFT) is applied to convert time signals into a frequency spectrum. This method is crucial for vibrodiagnostics because it enables precise fault localization by identifying discrete peaks at the characteristic frequencies of the bearing elements. This method is indispensable for accurate fault localization [9, 17].

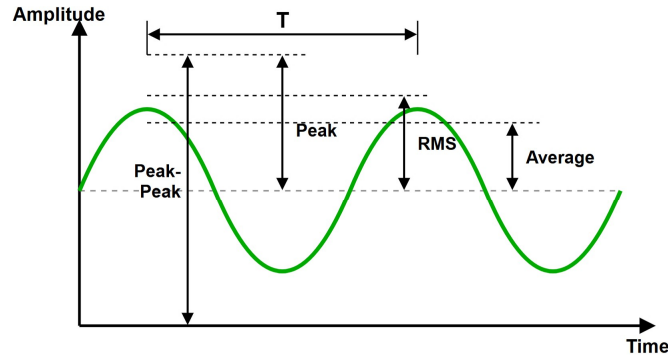
There are a large number of methods for analyzing vibration signals, which is why the choice of the appropriate technique is determined by the nature of the signal, the type of damage, and the operating conditions of the machine. Methods based on the Fourier transform are widely used in the diagnostics of rotating machines [18].

### 2.1 Vibration Analysis in the Time Domain

Measuring a time signal is the simplest form of vibration analysis. There are a number of characteristics for evaluating the amplitude (level) of a vibration signal (Figure 2):

- Peak value;
- Peak-to-peak value;

- Average value;
- RMS;
- Crest factor;
- Periodicity/Repetition rate;
- Duration.



**Figure 2.** Time domain signal characteristics

Note: RMS = root mean square.

The peak amplitude,  $A_z$ , is a parameter particularly useful for expressing the level of short-term shock vibrations. This parameter only indicates the maximum amplitude value, while it does not take into account the time history of the signal. The mean value,  $A_{sr}$ , is a parameter that takes into account the time history of the signal. The use of this parameter is of limited practical value, since it does not have a direct correlation with any physical quantity. The effective value of the signal,  $A_{ef}$ , is the most important measure of the vibration amplitude because it takes into account the time history of the signal. In this way, this parameter gives the signal amplitude a value that is directly related to the energy content of the signal, i.e., the destructive ability of a given vibration. For a harmonic periodic signal, the relations are:

$$A_{ef} = \frac{A_z}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}} A_{sr} \quad (1)$$

The quantities shown that describe the time signal do not only apply to a simple sinusoidal signal, but to all common vibration signals that can be obtained from machines, which are composed of many sinusoidal components.

## 2.2 Vibration Analysis in the Frequency Domain

Frequency domain signal analysis is the most important tool used for fault detection through vibration measurement and analysis. Figure 3 shows a comparative view of the change in the overall vibration level (Overall Level) and frequency spectrum (Frequency Spectrum) over time for two different machine systems: a fan (top) and a gearbox (bottom).

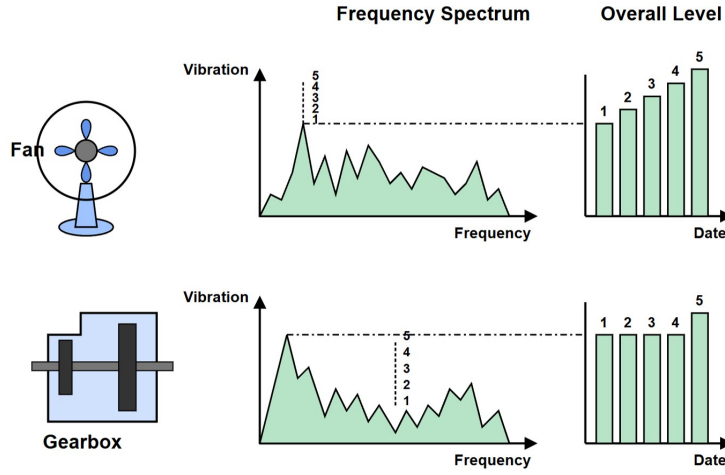
FFT is a method of converting signals recorded in the time domain into the frequency domain. By analyzing the frequency domain signal, it is possible to detect various faults.

After a critical analysis of the signal in the frequency domain, a diagnosis is made, and two basic rules must be followed:

- Each characteristic frequency in the spectrum must be associated with a certain part of the technical system or a physical quantity related to the execution of the process;
- Any characteristic amplitude and frequency cannot be changed without reason. Any change in the listed quantities indicates a change in the dynamic behavior of the technical system. Any change indicates a change in the state of the technical system.

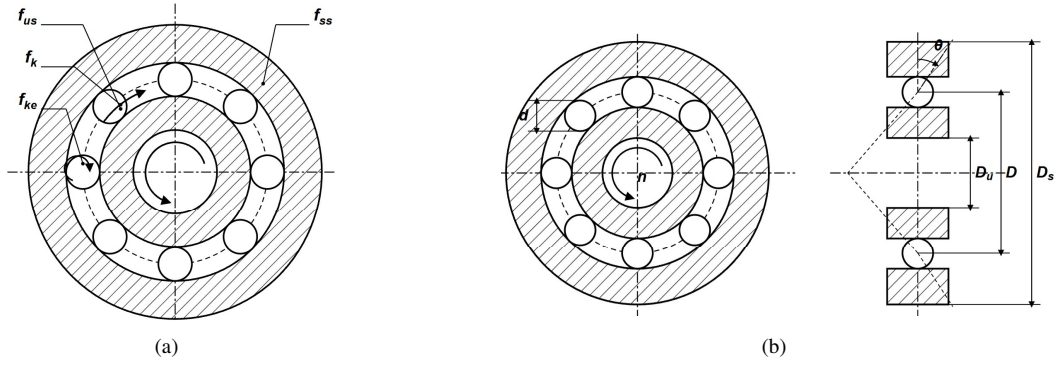
The process of developing damage to rolling bearings causes changes in the recorded vibration signals that manifest themselves in parts of the frequency spectrum, which correspond to the characteristic frequencies of rolling bearing vibrations. Therefore, if there is damage to any of the bearing elements, an increase in the intensity of the vibration amplitude will occur at the corresponding characteristic frequency. The characteristic frequencies (Figure 4a) of rolling bearings are the following:

- Characteristic frequency of the cage  $f_k$ .
- Characteristic frequency of the inner raceway  $f_{us}$ .
- Characteristic frequency of the outer raceway  $f_{ss}$ .
- Characteristic frequency of the rolling elements  $f_{ke}$ .



**Figure 3.** Examples of total vibration level measurement and frequency analysis

The characteristic vibration frequencies of rolling bearings are defined by the geometric relationships and kinematic parameters of the motion of the bearing elements, and therefore can be determined analytically. Figure 4b shows the quantities that will be used to determine the characteristic frequencies of rolling bearings.



**Figure 4.** Schematic representation of a rolling bearing: (a) bearing characteristic fault frequencies; (b) basic dimensions of the rolling bearing

The following expressions given are expressions for calculating the characteristic frequencies of a rolling bearing. Cage Fault Frequency:

$$f_k = \frac{1}{2} \left( 1 - \frac{d}{D} \cos \theta \right) \cdot f \quad (2)$$

where,  $f$  means shaft rotation frequency,  $d$  means rolling element diameter,  $D$  means bearing diameter, and  $\theta$  means contact angle.

The average diameter is calculated according to the formula:

$$D \approx \frac{D_u + D_s}{2} \quad (3)$$

where,  $D_u$  means inner diameter and  $D_s$  means outer diameter of the bearing.

The Ball Pass Frequency Inner Ring ( $f_{us}$ ) represents the frequency at which the rolling elements pass over a single point on the inner raceway, and it can be calculated using the following expression:

$$f_{us} = \frac{N}{2} \left( 1 + \frac{d}{D} \cos \theta \right) \cdot f \quad (4)$$

The Ball Pass Frequency Outer Ring ( $f_{ss}$ ) is the frequency at which the rolling elements pass over a single point on the outer raceway. This frequency can be calculated using the following expression:

$$f_{ss} = \frac{N}{2} \left( 1 - \frac{d}{D} \cos \theta \right) \cdot f \quad (5)$$



The Ball Spin Frequency (BSF) ( $f_{ke}$ ) represents the rotational frequency of an individual rolling element about its own axis, and it can be calculated using the following expression:

$$f_{ke} = \frac{d}{2D} \left[ 1 - \left( \frac{d}{D} \cos \theta \right)^2 \right] \cdot f \quad (6)$$

In cases where the exact bearing type and detailed geometric dimensions are unknown, the following simplified approximate formulas can be utilized to estimate the bearing characteristic fault frequencies:

$$f_k \approx 0.4 \cdot f; f_{ss} \approx 0.4 \cdot f \cdot N; f_{us} \approx 0.6 \cdot f \cdot N \quad (7)$$

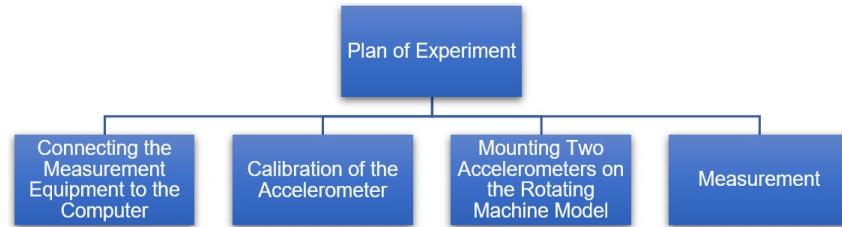
where,  $N$  means number of rolling elements.

Analytically calculated characteristic fault frequencies for rolling element bearings often differ from those found in measured vibration spectra. These differences are largely due to the following reasons:

- Differences in operating speed: Differences between the actual rotational speed and the nominal or declared speed of the system;
- Manufacturing tolerances and design changes: Changes in the internal geometry of bearings such as pitch diameter, rolling element size or number of elements that are made by manufacturers without prior notice;
- Axial loading effects: Large axial loads alter the contact angle and the pitch path of the rolling element;
- Geometric wear: Progressive wear of the raceways and rolling elements, increase of the effective pitch circumference;
- Kinematic slip: Pure sliding or skidding in the contact zones at high speeds and light loads, disturbing the theoretical kinematics.

### 2.3 Equipment for the Experiment

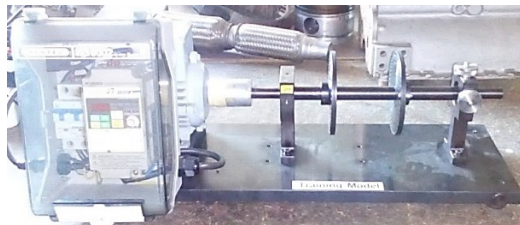
In this paper, an experimental study was conducted, which included familiarization with the measuring equipment and the model of the rotating machine, and identification of the characteristic frequencies of rolling bearings in the vibration signal. Figure 5 shows the experimental plan. The experiment was performed for 4 different frequency values, the transducers were placed in 2 places, the measurement for each frequency lasted 5 seconds, and the frequencies for which the measurement was performed were as follows: 20, 40, 60, and 80 Hz.



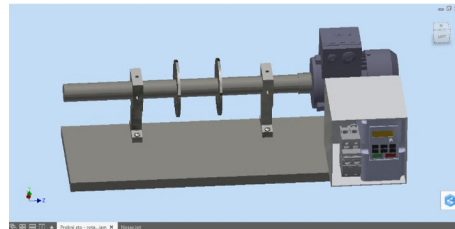
**Figure 5.** Experimental plan

For the purposes of conducting experimental research, the following equipment was used:

- A model of a rotary machine, which served as a physical model for simulating operating modes (Figure 6);



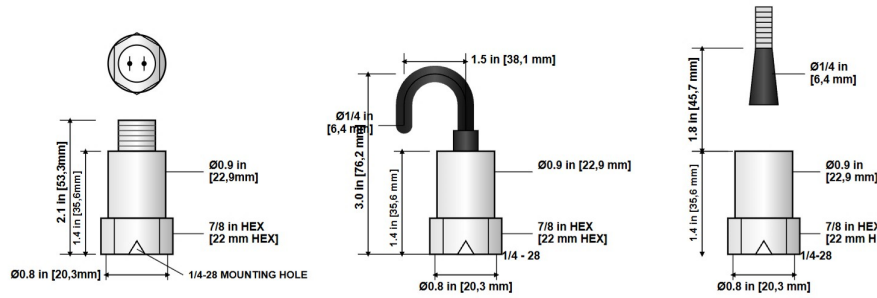
(a)



(b)

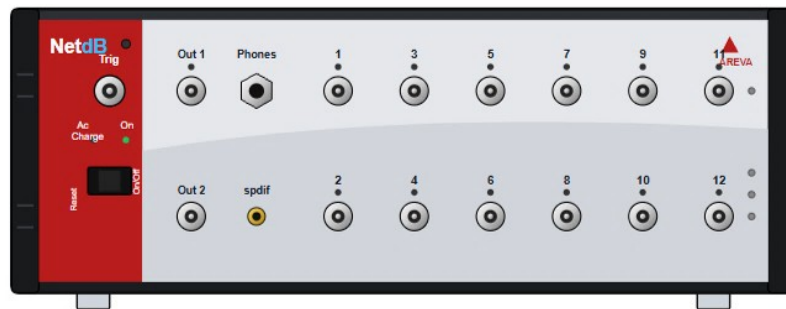
**Figure 6.** Rotary machine table: (a) picture of table; (b) 3D model

- Piezoelectric acceleration sensors (accelerometers), set up to measure the vibration response (Figure 7);
- NetdB12, a multi-channel device for continuous signal acquisition and digitization (Figure 8);
- A computer with an accompanying software package for processing and spectral analysis of measured signals.



**Figure 7.** Accelerometers

Note: HEX = hexagonal (across flats).



**Figure 8.** Schematic representation of the NetdB12 data acquisition unit

When conducting the experiment, the accelerometers were positioned in the radial directions and mounted directly on the bearing housing to ensure effective transmission of vibration signals from the bearing. This sensor arrangement followed recommendations from the relevant vibrodiagnostic literature and was selected to provide reliable detection of vibration components associated with bearing condition. Consistent sensor positioning was maintained throughout the experiment to improve the repeatability and comparability of the measured signals.

It is crucial to place the sensor as close as possible to the vibration source, as positioning it elsewhere leads to signal attenuation or unwanted amplification caused by structural oscillations. Measuring at locations distant from the primary source introduces parasitic dynamics that distort the acquired data. This methodology aligns with the approach in their research on diagnostics through non-invasive measurements [16, 19]. They point out that vibration signals reliably reflect internal phenomena only when acquired from external surfaces in direct mechanical coupling with the source such as the cylinder head in their studies, or the bearing housing in this work. Direct positioning ensures that the vibration signal remains an accurate indicator of internal process dynamics without interference from structural natural frequencies.

The test bench is powered by a standard 3-phase asynchronous motor with a squirrel-cage rotor. Electric motors of this type are widely used to drive rotating machines in industry. The inverter that the test bench has the ability to adjust many parameters. The most important parameters from the point of view of application on the test bench for complex rotor diagnostics are: maximum frequency (80 Hz), start-up time and deceleration time. The design of the test bench is such that it can provide simulation of factors that lead to failures such as: unbalance, the appearance and development of cracks and misalignment.

Characteristics of the acquisition device:

- Personal computer connection: Internet cable RJ45 100 Mbits, 2 universal serial bus ports in the back personal digital assistant port and WiFi 802.11 g 54 Mbits;
- Data storage: 100 GB hard disk drive (12 hours/continuous measurement with 12 channels at 51.2 kHz 24 bit);
- Connectivity: 12 channels that can be connected up to 51.2 kHz 24 bit on each channel of the NetdB-DAQ12 unit, as shown in Figure 9.

Before the measurement itself, it is necessary to calibrate the sensor to be used. The calibration was performed using the aforementioned software. The calibration device and the acceleration sensor calibration are shown in Figure 10. The characteristic of the acceleration sensor calibration device is that at a frequency of 159.2 Hz, it gives an acceleration value of 1 g, which can be seen from Figure 10.

The bearing installed in the test bench, i.e., the bearing tested, is a bearing with the designation 6003ZZ. The dimensions of the bearing and the 3D model of the bearing are shown in Figure 11.

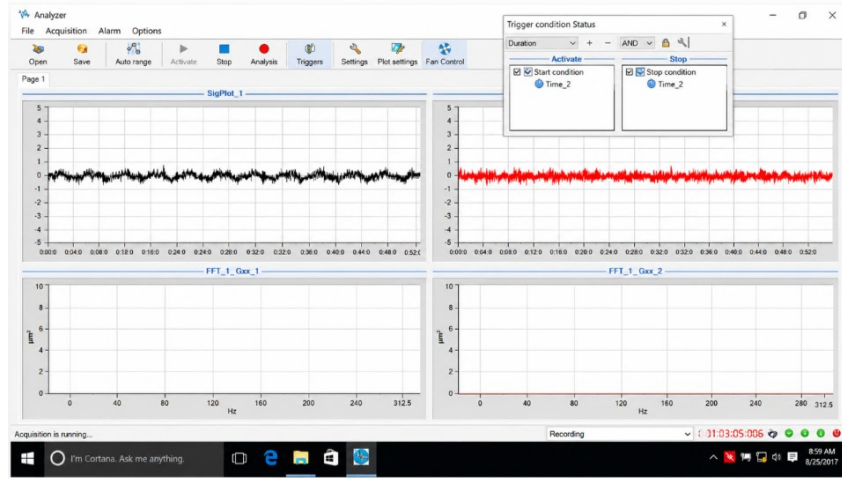


Figure 9. Acquisition software window screenshot

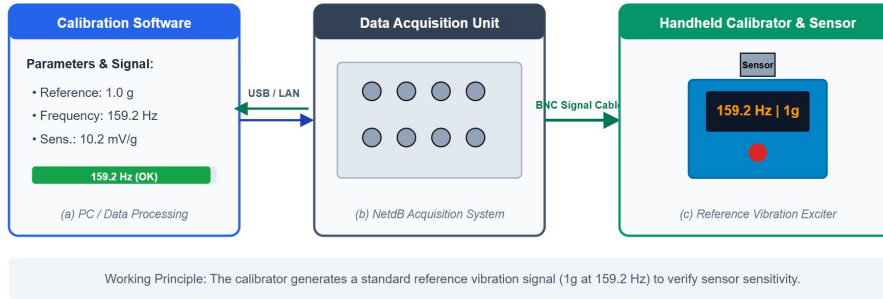


Figure 10. Calibration

Note: USB = universal serial bus; LAN = local area network; BNC = Bayonet Neill–Concelman; PC = personal computer; Sens. = sensitivity.

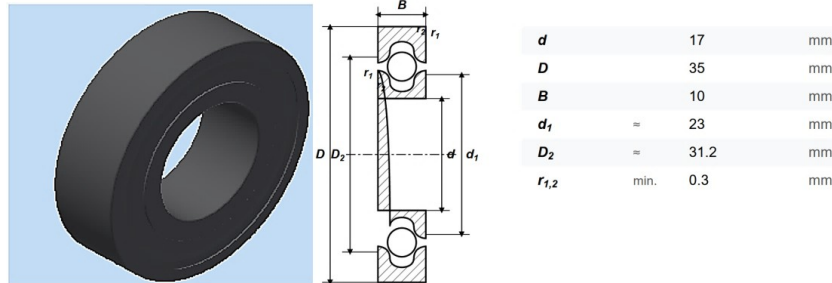


Figure 11. Bearing 6003 2ZR

### 3 Results and Discussion

According to Du et al. [20], recent studies confirm that vibration analysis remains the primary technique for rolling bearing fault diagnosis. Although advanced time-frequency and machine-learning methods have been introduced for variable-speed applications, the identification of characteristic fault frequencies through Fourier-based analysis continues to provide a reliable and physically interpretable foundation for bearing condition monitoring.

The experiment was conducted across four distinct inverter operating frequencies: 20, 40, 60, and 80 Hz. Two vibration sensors were positioned on the non-drive-end bearing in the radial directions (vertical and horizontal). Data acquisition lasted 5 seconds for each frequency step.

The diagrams shown below represent the vibration spectrum in the time and frequency domains. The signal was recorded using the equipment described in the previous section, then the values obtained during the experiment were further processed in the program “Origin Pro 2025” in order to obtain the diagrams.

The diagrams shown below show peaks that deviate from the rest of the spectrum of the recorded signal and the peaks can occur as a result of: the natural frequencies of the test table, imbalance, rotating parts, misalignment, insufficient attachment to the base or a bad and damaged bearing.



Also shown are diagrams indicating the characteristic frequencies of the outer raceway  $f_{ss}$ . The values calculated for the frequencies of the outer raceway and read from the diagrams for the frequencies of the outer raceway are as follows:

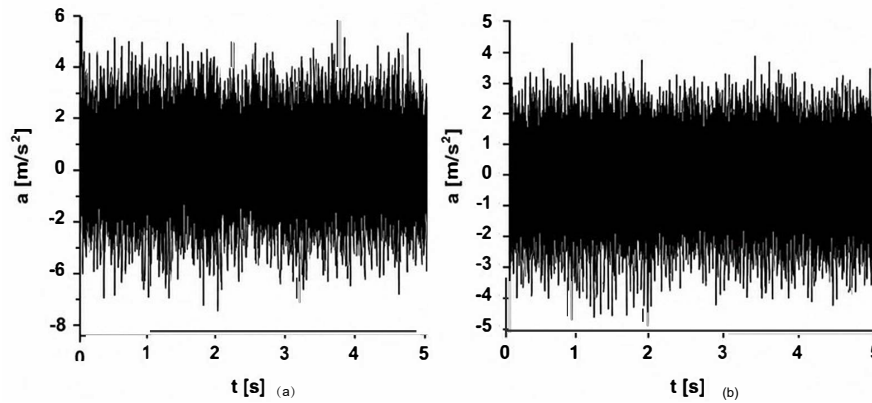
- At a frequency of 20 Hz,  $f_{ss} = 87.1$  Hz was calculated, while the value read on the diagram was 100.15493 Hz.
- At a frequency of 40 Hz, the calculation results for  $f_{ss} = 174.2$  Hz, while the value read on the diagram is 200.30986 Hz.
- At a frequency of 60 Hz, the calculation results for  $f_{ss} = 261.3$  Hz, while the value read on the diagram is 260.40154 Hz.
- At a frequency of 80 Hz, the calculation results for  $f_{ss} = 348.4$  Hz, while the value read on the diagram is 340.52592 Hz.

The characteristic frequencies on the diagrams deviate slightly from the values obtained mathematically. The reasons for the possible deviation are given in Chapter 2.2 of this paper.

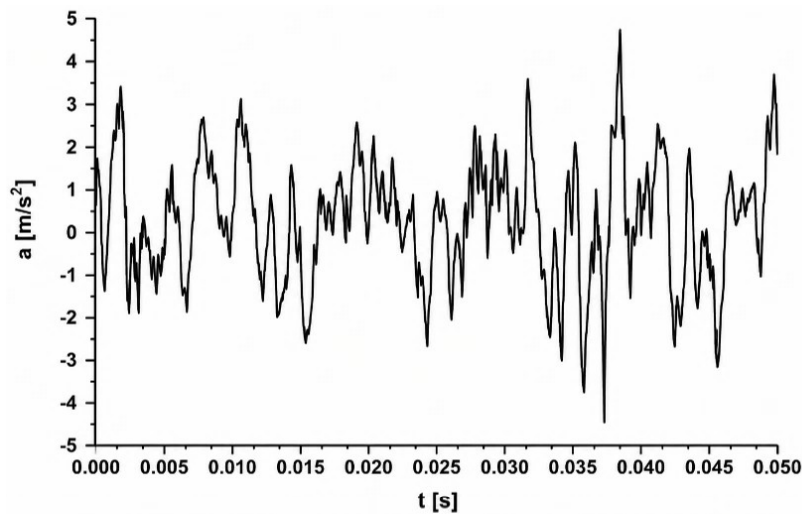
### 3.1 Results for a Frequency of 20 Hz

For the first experimental mode, the shaft rotation frequency was set to 20 Hz. Vibration signals were acquired simultaneously with two accelerometers placed in radial directions.

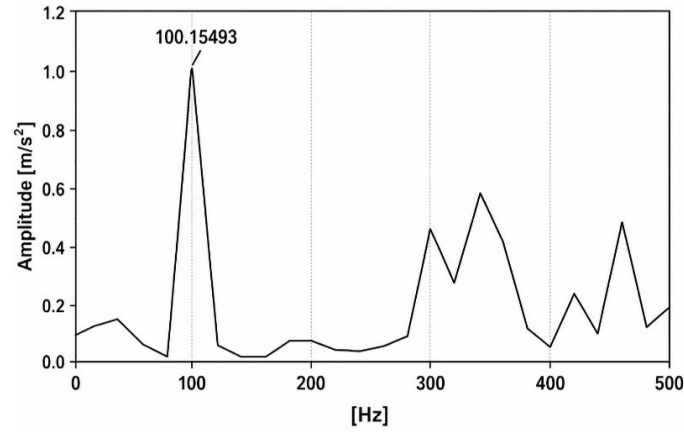
The time diagrams (Figure 12a and 12b) show a characteristic noise with occasional pulse peaks, which is shown in more detail in the signal segment up to 0.05 s (Figure 13). Spectrum analysis identified a dominant peak at a frequency of 100.15493 Hz (Figure 14).



**Figure 12.** Measured acceleration time-domain signals: (a) time diagram for the first signal; (b) time diagram for the second signal (shaft rotation frequency = 20 Hz)



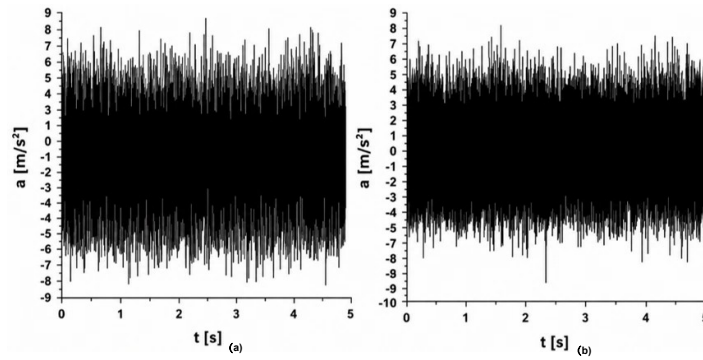
**Figure 13.** Time diagram—segment up to 0.05 s (shaft rotation frequency = 20 Hz)



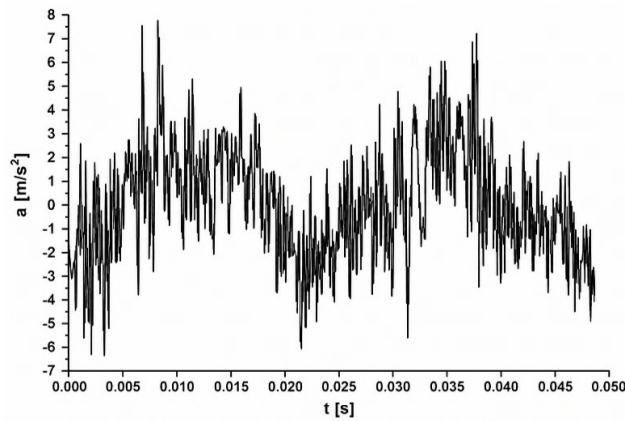
**Figure 14.** Fast Fourier transform diagram with characteristic frequency indicated (shaft rotation frequency = 20 Hz)

### 3.2 Results for a Frequency of 40 Hz

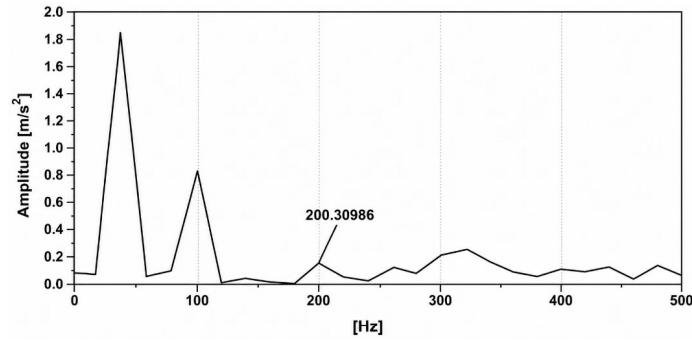
In the second experiment, the shaft rotation frequency was set to 40 Hz, and vibration signals were simultaneously acquired through two NetdB12 channels using radially mounted accelerometers on the bearing housing. The time diagrams (Figure 15a and 15b) show periodic pulse amplitude changes, which is shown in more detail in the signal segment up to 0.05 s (Figure 16). These periodic shocks are a direct consequence of the rolling elements passing over localized damage, which, according to the literature, causes sudden changes in contact stresses and generates short-duration pulses. Spectrum analysis identified a dominant peak at a frequency of 200.30986 Hz (Figure 17).



**Figure 15.** Measured acceleration time-domain signals: (a) time diagram for the first signal; (b) time diagram for the second signal (shaft rotation frequency = 40 Hz)



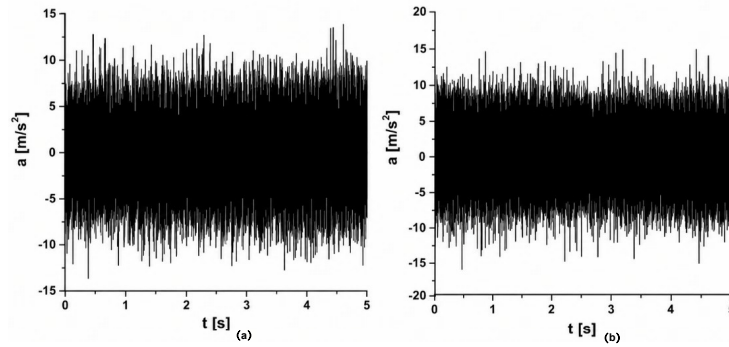
**Figure 16.** Time diagram—segment up to 0.05 s (shaft rotation frequency = 40 Hz)



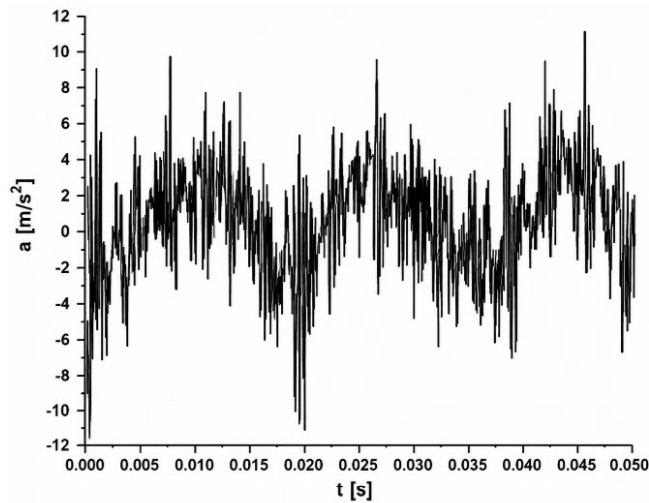
**Figure 17.** Fast Fourier transform diagram with characteristic frequency indicated (shaft rotation frequency = 40 Hz)

### 3.3 Results for a Frequency of 60 Hz

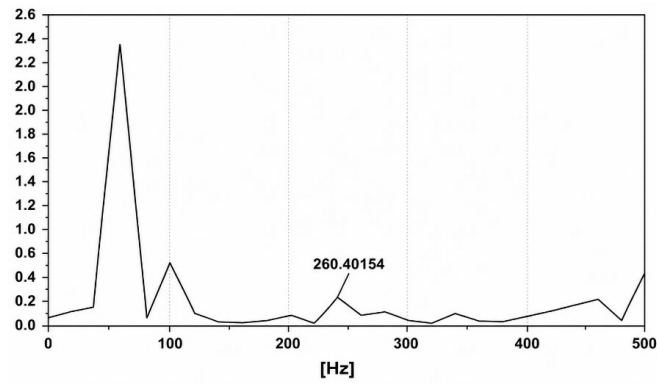
In the third experimental mode, the shaft rotation frequency was set to 60 Hz. Vibration signals were acquired using the NetdB12 system. The visualization of the signal in the time domain (Figure 18a and 18b) shows significantly higher vibration amplitudes compared to the previous modes of 20 Hz and 40 Hz, which is a direct consequence of the higher kinetic energy of the rotating masses. In the detailed display of the signal segment up to 0.05 s (Figure 19), denser periodic pulse shocks are observed, which corresponds to a higher frequency of rolling element passage over localized damage at this speed. In this case, a pronounced dominant peak at a frequency of 260.40154 Hz was identified (Figure 20).



**Figure 18.** Measured acceleration time-domain signals: (a) time diagram for the first signal; (b) time diagram for the second signal (shaft rotation frequency = 60 Hz)



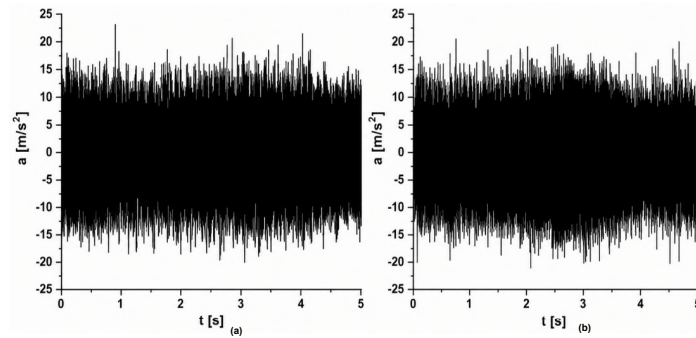
**Figure 19.** Time diagram—segment up to 0.05 s (shaft rotation frequency = 60 Hz)



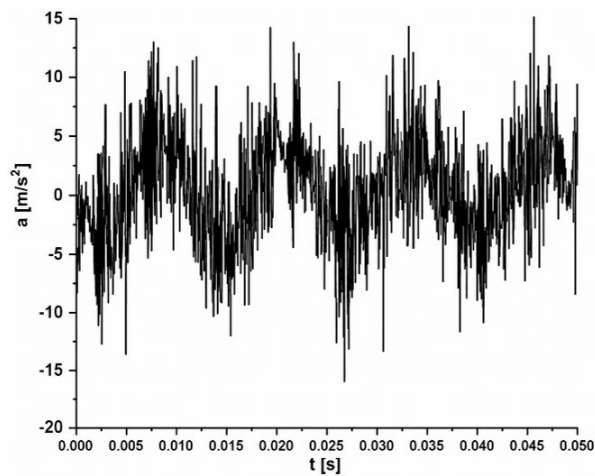
**Figure 20.** Fast Fourier transform diagram with characteristic frequency indicated (shaft rotation frequency = 60 Hz)

### 3.4 Results for a Frequency of 80 Hz

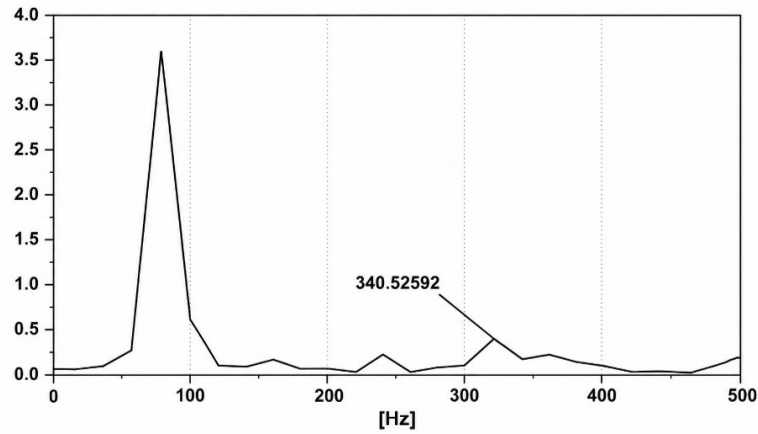
The fourth and most intense experimental mode was carried out at a shaft rotation frequency of 80 Hz, which is the maximum speed predicted by this setting. The signals were acquired via the NetdB12 system with two accelerometers in radial directions. The visualization in the time domain (Figure 21a and 21b) clearly shows significantly higher acceleration amplitudes compared to the previous modes, which is a consequence of the high centrifugal energy at 4800 rpm. In the enlarged signal segment (Figure 22), pronounced periodic high-energy shocks are observed, which are caused by very fast passages of the rolling elements over the damage zone. Spectrum analysis identified a dominant diagnostic peak at a frequency of 340.52592 Hz (Figure 23).



**Figure 21.** Measured acceleration time-domain signals: (a) time diagram for the first signal; (b) time diagram for the second signal (shaft rotation frequency = 80 Hz)



**Figure 22.** Time diagram—segment up to 0.05 s (shaft rotation frequency = 80 Hz)



**Figure 23.** Fast Fourier transform diagram with characteristic frequency indicated (shaft rotation frequency = 80 Hz)

### 3.5 Analytical Determination of Characteristic Frequency

Early detection of rolling bearing damage is based on the identification of specific frequency components in the vibration spectrum, which are directly related to the kinematics of the motion of the bearing elements. The characteristic vibration frequencies are defined by geometric relationships and kinematic parameters, and for the case of rotating inner ring with stationary outer ring, they can be determined analytically.

For the 6003 2ZR bearing, the necessary data for calculating the characteristic frequencies are listed in Table 1.

**Table 1.** Value for calculation

| Parameter                  | Symbol     | Value  | Unit |
|----------------------------|------------|--------|------|
| Inner diameter             | $(D_u)$    | 17     | mm   |
| Outer diameter             | $(D_s)$    | 35     | mm   |
| Rolling element diameter   | $(d)$      | 4.7621 | mm   |
| Number of rolling elements | $(N)$      | 10     | —    |
| Contact angle              | $(\theta)$ | 45     | °    |

Note: — indicates not applicable.

The Fourier Transform represents one of the most common and fundamental signal processing techniques used in vibration-based fault diagnosis, which enables the identification of characteristic fault frequencies in the frequency domain. These methods continue to provide reliable diagnostic information for rotating machinery. Typical bearing faults include outer race damage, inner race damage, rolling element faults and cage faults. Diagnoses are usually performed based on vibration measurements, where the primary diagnostic indicators consist of characteristic defect frequencies and time-domain statistical characteristics [21, 22].

Using expressions from Section 2, the characteristic frequencies are determined.

The average diameter is calculated according to the formula:

$$D \approx \frac{D_u + D_s}{2} = \frac{17 + 35}{2} = \boxed{26 \text{ mm}} \quad (8)$$

Cage Characteristic Frequency—Fundamental Train Frequency:

$$20 \text{ Hz} : f_k = \frac{1}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{1}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 20 = \frac{1}{2} \cdot 0.871 \cdot 20 = \boxed{8.71 \text{ Hz}} \quad (9)$$

$$40 \text{ Hz} : f_k = \frac{1}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{1}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 40 = \frac{1}{2} \cdot 0.871 \cdot 40 = \boxed{17.42 \text{ Hz}} \quad (10)$$

$$60 \text{ Hz} : f_k = \frac{1}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{1}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 60 = \frac{1}{2} \cdot 0.871 \cdot 60 = \boxed{26.13 \text{ Hz}} \quad (11)$$

$$80 \text{ Hz} : f_k = \frac{1}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{1}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 80 = \frac{1}{2} \cdot 0.871 \cdot 80 = \boxed{34.84 \text{ Hz}} \quad (12)$$



Ball Pass Frequency of the inner Race (BPFI)—Characteristic Frequency of the Inner Race:

$$20 \text{ Hz} : f_{us} = \frac{N}{2} \left( 1 + \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 + \frac{4.762}{26} \cos 45^\circ \right) \cdot 20 = 5 \cdot 1.129 \cdot 20 = \boxed{112.9 \text{ Hz}} \quad (13)$$

$$40 \text{ Hz} : f_{us} = \frac{N}{2} \left( 1 + \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 + \frac{4.762}{26} \cos 45^\circ \right) \cdot 40 = 5 \cdot 1.129 \cdot 40 = \boxed{225.8 \text{ Hz}} \quad (14)$$

$$60 \text{ Hz} : f_{us} = \frac{N}{2} \left( 1 + \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 + \frac{4.762}{26} \cos 45^\circ \right) \cdot 60 = 5 \cdot 1.129 \cdot 60 = \boxed{338.7 \text{ Hz}} \quad (15)$$

$$80 \text{ Hz} : f_{us} = \frac{N}{2} \left( 1 + \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 + \frac{4.762}{26} \cos 45^\circ \right) \cdot 80 = 5 \cdot 1.129 \cdot 80 = \boxed{451.6 \text{ Hz}} \quad (16)$$

Ball Pass Frequency of the Outer Race (BPFO)—Characteristic Frequency of the Outer Race:

$$20 \text{ Hz} : f_{ss} = \frac{N}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 20 = 5 \cdot 0.871 \cdot 20 = \boxed{87.1 \text{ Hz}} \quad (17)$$

$$40 \text{ Hz} : f_{ss} = \frac{N}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 40 = 5 \cdot 0.871 \cdot 40 = \boxed{174.2 \text{ Hz}} \quad (18)$$

$$60 \text{ Hz} : f_{ss} = \frac{N}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 60 = 5 \cdot 0.871 \cdot 60 = \boxed{261.3 \text{ Hz}} \quad (19)$$

$$80 \text{ Hz} : f_{ss} = \frac{N}{2} \left( 1 - \frac{d}{D} \cos 45^\circ \right) \cdot f = \frac{10}{2} \left( 1 - \frac{4.762}{26} \cos 45^\circ \right) \cdot 80 = 5 \cdot 0.871 \cdot 80 = \boxed{348.4 \text{ Hz}} \quad (20)$$

BSF—Characteristic Frequency of the Rolling Element:

$$20 \text{ Hz} : f_{ke} = \frac{d}{2D} \left[ 1 - \left( \frac{d}{D} \cos 45^\circ \right)^2 \right] \cdot f = \frac{4.762}{2 \cdot 26} \left[ 1 - \left( \frac{4.762}{26} \cos 45^\circ \right)^2 \right] \cdot 20 \quad (21)$$

$$f_{ke} = 0.092 \cdot 0.9834 \cdot 20 = \boxed{1.809 \text{ Hz}} \quad (22)$$

$$40 \text{ Hz} : f_{ke} = \frac{d}{2D} \left[ 1 - \left( \frac{d}{D} \cos 45^\circ \right)^2 \right] \cdot f = \frac{4.762}{2 \cdot 26} \left[ 1 - \left( \frac{4.762}{26} \cos 45^\circ \right)^2 \right] \cdot 40 \quad (23)$$

$$f_{ke} = 0.092 \cdot 0.9834 \cdot 40 = \boxed{3.619 \text{ Hz}} \quad (24)$$

$$60 \text{ Hz} : f_{ke} = \frac{d}{2D} \left[ 1 - \left( \frac{d}{D} \cos 45^\circ \right)^2 \right] \cdot f = \frac{4.762}{2 \cdot 26} \left[ 1 - \left( \frac{4.762}{26} \cos 45^\circ \right)^2 \right] \cdot 60 \quad (25)$$

$$f_{ke} = 0.092 \cdot 0.9834 \cdot 60 = \boxed{5.428 \text{ Hz}} \quad (26)$$

$$80 \text{ Hz} : f_{ke} = \frac{d}{2D} \left[ 1 - \left( \frac{d}{D} \cos 45^\circ \right)^2 \right] \cdot f = \frac{4.762}{2 \cdot 26} \left[ 1 - \left( \frac{4.762}{26} \cos 45^\circ \right)^2 \right] \cdot 80 \quad (27)$$

$$f_{ke} = 0.092 \cdot 0.9834 \cdot 80 = \boxed{7.238 \text{ Hz}} \quad (28)$$

## 4 Discussion

Vibration analysis represents the most widely adopted technique for condition monitoring of rotating machinery. Mechanical damage generates vibration responses, which can be captured prior to catastrophic failure. Accelerometers serve as standard transducers for vibration measurement and are mounted at appropriate locations on machines to reliably monitor the technical state of their components. Vibration analysis enables the detection of various faults in rotating machinery; owing to its satisfactory reliability, it has become one of the prevailing condition monitoring techniques in industrial practice [23].

Early identification of incipient bearing defects is critical to prevent unexpected shutdowns of rotating equipment. Frequency-domain analysis of vibration signals facilitates an improved understanding of the evolution of bearing faults, where distinct frequency bands often exhibit early indicators of component degradation. Any undetected and unresolved minor defect may evolve into catastrophic failure, eventually leading to the shutdown of the entire mechanical system [24].

Vibration diagnostic techniques enable the detection of mechanical variations occurring during the operation of mechanical systems, supporting fault root cause elimination and the prevention of equipment failures. As a

conventional preventive maintenance tool, vibration diagnostics can substantially reduce maintenance expenditure while improving the safety, reliability and service life of mechanical systems.

In the bearing tests, vibration frequency spectra and time-domain waveforms were recorded during operation. Through the assessment of measured signals, bearing conditions and potential mechanical damage can be identified. Machine faults are diagnosed according to dominant frequency components in spectral data and waveform abnormalities within time-domain records.

Each mechanical fault generates vibration components at specific frequencies, and the amplitude corresponding to a characteristic frequency is proportional to fault severity. Therefore, amplitude–frequency vibration characteristics can be used to determine fault types (via vibration frequency) and fault severity (via associated vibration amplitude).

Frequency analysis constitutes an effective diagnostic tool for evaluating the health state of mechanical systems. Within the vibration frequency spectrum of rotating machines, prominent spectral peaks appear at specific frequencies. Machine or system conditions can be evaluated by analysing these peaks and matching them against theoretical characteristic frequencies of rotating components.

Variations in characteristic amplitudes and frequencies do not occur randomly. Any observable change reflects altered dynamic behaviour and evolving health conditions of the mechanical system.

In this work, experiments were carried out under four different excitation frequencies of the frequency controller. Two vibration sensors were arranged along radial measurement directions (vertical and horizontal). Measurements lasted 5 seconds for each excitation frequency of the regulator, with test frequencies set to 20, 40, 60, and 80 Hz.

Spectrograms from experimental results display prominent peaks that stand out from the background spectrum of acquired signals. Such peaks may originate from multiple sources: natural frequencies of the test rig, rotor unbalance, rotating components, shaft misalignment, insufficient fixture stiffness, or defective and damaged bearings.

A core part of this discussion concerns the comparison between analytically calculated bearing characteristic frequencies ( $f_{ss}$ ) and frequency components extracted via FFT. Experimental spectral peaks (e.g., 101.15 Hz under 20 Hz operation and 200.31 Hz under 40 Hz operation) deviate from corresponding theoretical values (87.1 Hz and 174.2 Hz, respectively). In academic research, such discrepancies are not regarded as measurement errors but as outcomes of real bearing physical behaviour. Ideal kinematic equations assume pure rolling motion, whereas rolling element skidding and sliding inevitably exist under practical operating conditions, leading to shifts of actual defect frequencies in vibration spectra.

For damaged bearings, vibration amplitudes rise in the time domain. Meanwhile, characteristic frequencies corresponding to specific damage types emerge in frequency-domain spectra, which provides the theoretical foundation for bearing fault diagnosis [25].

Experimental results (Table 2) indicate that the consistency between analytical and measured frequencies depends on rotational speed. The minimum deviation is achieved at 60 Hz, where vibration signals exhibit optimal stability and characteristic frequencies appear as distinct spectral peaks. By contrast, larger deviations at 20 Hz can be explained by weaker excitation energy, lower signal-to-noise ratio and stronger interference from background vibration. At 80 Hz, frequency deviations moderately increase due to rolling element slip, centrifugal effects and dynamic variations induced by higher rotational speeds. The overall deviations fall within reasonable ranges expected for practical working conditions, verifying the effectiveness of FFT-based vibration analysis for bearing fault identification.

**Table 2.** Results of experiment measurement

| Frequency | BPFO  | Fast Fourier Transform | Deviation |
|-----------|-------|------------------------|-----------|
| 20 Hz     | 87.1  | 101.15                 | ≈14.05 Hz |
| 40 Hz     | 174.2 | 200.31                 | ≈26.11 Hz |
| 60 Hz     | 261.3 | 260.40                 | ≈0.90 Hz  |
| 80 Hz     | 348.4 | 340.53                 | ≈7.87 Hz  |

Note: BPFO = ball pass frequency of the outer race.

## 5 Conclusions

Conducted experimental research on a rotary machine model with a built-in bearing 6003 2ZR confirmed that vibration analysis is a highly reliable method for early identification and localization of damage. The used measuring chain, with the NetdB12 acquisition system and piezoelectric accelerometers, enabled precise monitoring of the bearing dynamics in real time.

Based on the analysis of the obtained results, the following key conclusions can be drawn:

- Validity of frequency analysis: Application of FFT proved to be the most effective diagnostic tool, as it clearly identified the peaks corresponding to bearing element damage at all tested modes (20, 40, 60, and 80 Hz).

- Accuracy of analytical models: Experimentally determined frequencies showed a high degree of agreement with theoretical calculations. The 60 Hz mode is particularly noteworthy, where the deviation was minimal (only 0.9 Hz), while the variations at other frequencies remained within the limits that are natural for the real operating conditions of machine systems.
- Significance of the measurement site: Placing the sensor directly on the bearing housing in radial directions proved to be crucial for obtaining a clean signal without significant attenuation, which allowed a clear distinction between the useful fault signal and the background noise.
- Condition Indicators: The value of the signal has been confirmed to increase with increasing vibration intensity and rotational speed, serving as a direct indicator of the failure energy and general mechanical condition of the bearing.

The robustness of the proposed diagnostic approach under various operational conditions such as different rotational speeds and load levels should be investigated in future work. Further, advanced signal processing techniques such as time–frequency analysis and machine learning methods and method based on artificial intelligence can be explored and compared to the traditional time domain and FFT based methods used in this study. Finally, further experimental validation using different bearing types and different levels of fault severity would add additional support to the applicability of the proposed methodology for industrial condition monitoring and predictive maintenance.

This work represents a solid basis for the implementation of preventive maintenance, thus enabling the extension of the working life of machines and the prevention of sudden stoppages in industrial processes.

### Author Contributions

Conceptualization, N.K.; methodology, N.K.; validation, D.M.; formal analysis, N.K.; investigation, N.K.; resources, N.K.; data curation, D.M.; writing—original draft preparation, N.K.; writing—review and editing, D.M. and N.K.; visualization, N.K.; supervision, D.M. All authors have read and agreed to the published version of the manuscript.

### Data Availability

The data used to support the research findings are available from the corresponding author upon request.

### Conflicts of Interest

The authors declare no conflicts of interest.

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